Practice 1 — Set Theory

Solutions to Selected Problems

Problems 2, 7, 11, 16, 20, 54 (3, 11), 58, 60

Discrete Mathematics — QYSU

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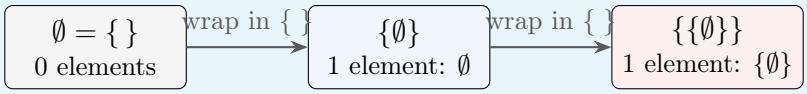
1 Problem 2 — Membership with $\emptyset$ and $\{\emptyset\}$

Theory Recap: Membership ($\in$) vs. Subset ($\subseteq$)

These two relations are fundamentally different:

- $a \in S$ means "a is an **element** of S."
- $A \subseteq S$ means "every element of A is also in S."

The empty set $\emptyset$ has no elements. The set $\{\emptyset\}$ is **not** empty—it has one element, namely $\emptyset$ itself. To decide "$x \in S$?" list the elements of S and check whether x is one of them.



Each wrapping step creates a new set whose sole element is the previous set.

Problem Statement

Which of the following statements are true?

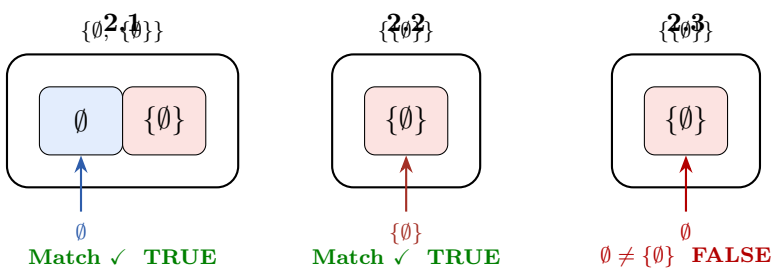
2.1. $\emptyset \in \{\emptyset,\ \{\emptyset\}\}$

2.2. $\{\emptyset\} \in \{\{\emptyset\}\}$

2.3. $\emptyset \in \{\{\emptyset\}\}$

Solution

The strategy is always the same: *list the elements* of the outer set, then check whether the object on the left of $\in$ matches any of them.



2.1. $\emptyset \in \{\emptyset,\ \{\emptyset\}\}$ — **TRUE.** The set has two elements: $\emptyset$ and $\{\emptyset\}$. The object $\emptyset$ is explicitly one of them.

2.2. $\{\emptyset\} \in \{\{\emptyset\}\}$ — **TRUE.** The set $\{\{\emptyset\}\}$ has exactly one element: $\{\emptyset\}$. The object on the left is $\{\emptyset\}$, which matches.

2.3. $\emptyset \in \{\{\emptyset\}\}$ — **FALSE.** The only element is $\{\emptyset\}$. We need $\emptyset = \{\emptyset\}$, but $|\emptyset| = 0$ while $|\{\emptyset\}| = 1$, so they are not equal.

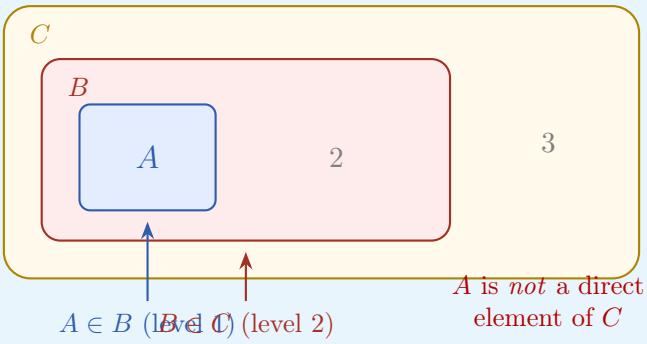
2.1 — True 2.2 — True 2.3 — False

2 Problem 7 — $\in$ is Not Transitive

Theory Recap: $\in$ is Not Transitive

$\subseteq$ **is** transitive: $A \subseteq B \wedge B \subseteq C \implies A \subseteq C$.
$\in$ is **not**: $A \in B \wedge B \in C \not\Longrightarrow A \in C$.

Why? $\in$ crosses one "level" of nesting. Two $\in$-steps cross *two* levels, so A is buried inside C but is not a direct element of C.



Problem Statement

Give examples of sets A, B, C such that $A \in B$, $B \in C$, but $A \notin C$.

Solution

Let

$$A = 1, \qquad B = \{1, 2\}, \qquad C = \big\{\{1, 2\},\ 3\big\}.$$

Condition	Check	Result
$A \in B$	Is $1 \in \{1, 2\}$?	Yes ✓
$B \in C$	Is $\{1, 2\} \in \big\{\{1, 2\},\ 3\big\}$?	Yes ✓
$A \notin C$	Is $1 \in \big\{\{1, 2\},\ 3\big\}$? The elements are $\{1, 2\}$ and 3.	$1 \notin C$ ✓

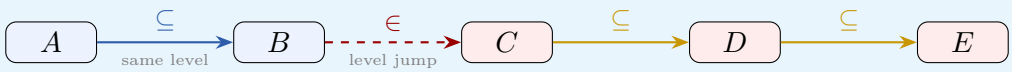
Key insight. The number 1 is an element of B, and B is an element of C, but the elements of C are $\{1, 2\}$ (a set) and 3 (a number). The number 1 is neither of these, so $1 \notin C$.

3 Problem 11 — Mixing $\in$ and $\subseteq$

Theory Recap: Mixing $\in$ and $\subseteq$

The two relations operate at different "levels":

- $A \subseteq B$: both sit at the same level—every element of A is also an element of B.
- $B \in C$: B itself becomes an object inside C—a jump *up* one level.



Problem Statement

Give examples of sets A, B, C, D, E satisfying:

$$A \subseteq B, \quad B \in C, \quad C \subseteq D, \quad D \subseteq E.$$

Solution

$$A = \{1\}, \quad B = \{1, 2\}, \quad C = \big\{\{1, 2\},\ 3\big\}, \quad D = \big\{\{1, 2\},\ 3,\ 4\big\}, \quad E = \big\{\{1, 2\},\ 3,\ 4,\ 5\big\}$$

Condition	Type	Verification
$A \subseteq B$	subset	Every element of $\{1\}$ is in $\{1, 2\}$: $1 \in \{1, 2\}$ ✓
$B \in C$	membership	$\{1, 2\}$ is listed as an element of $\big\{\{1, 2\}, 3\big\}$ ✓
$C \subseteq D$	subset	Elements $\{1, 2\}$ and 3 of C are both in D ✓
$D \subseteq E$	subset	Elements $\{1, 2\}$, 3, 4 of D are all in E ✓

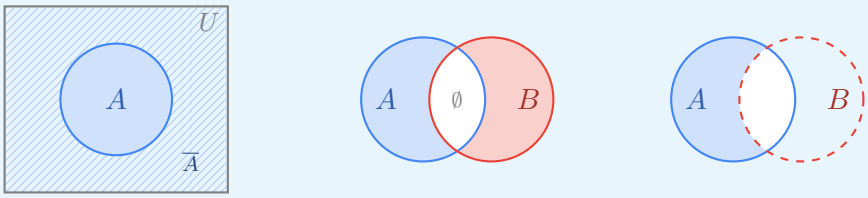
Why this works. $\subseteq$ preserves the "level": A and B contain the same kind of objects (numbers). The $\in$-step promotes B to an object inside C, after which C, D, E all live at the higher level (their elements include sets like $\{1, 2\}$).

4 Problem 16 — Describing Sets Under Operations

Theory Recap: Set Operations

For subsets of a universal set U:

Operation	Symbol	Definition
Complement	$\overline{A}$	$\{x \in U \mid x \notin A\}$
Set difference	$A \setminus B$	$\{x \in A \mid x \notin B\}$
Symmetric diff.	$A \triangle B$	$(A \setminus B) \cup (B \setminus A)$ = elements in exactly one of A, B



Complement $\quad A \triangle B \quad A \setminus B$

Problem Statement

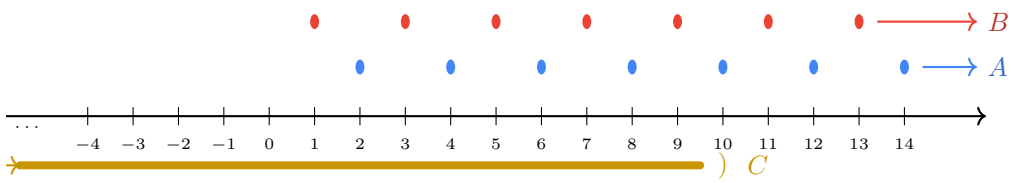
Let $\mathbb{Z}$ be the universal set. Define:

$$A = \{x \in \mathbb{Z} \mid \exists\, y \in \mathbb{Z}^+,\ x = 2y\} \qquad \text{(positive even integers: } 2, 4, 6, 8, \ldots)$$
$$B = \{x \in \mathbb{Z} \mid \exists\, y \in \mathbb{Z}^+,\ x = 2y - 1\} \qquad \text{(positive odd integers: } 1, 3, 5, 7, \ldots)$$
$$C = \{x \in \mathbb{Z} \mid x < 10\}$$

Describe: $\overline{A}$, $A \triangle B$, $\overline{C}$, $A \setminus C$, $C \setminus (A \triangle B)$.

Solution

Visualising the sets on the number line.



(a) $\overline{A}$**:** Everything in $\mathbb{Z}$ that is *not* a positive even integer:

$$\boxed{\overline{A} = \{\ldots, -2, -1, 0\} \cup \{1, 3, 5, 7, \ldots\} = \{x \in \mathbb{Z} \mid x \leq 0 \text{ or } x \text{ is odd}\}}$$

(b) $A \triangle B$**:** Since $A \cap B = \emptyset$ (no integer is both positive-even and positive-odd):

$$A \triangle B = A \cup B = \{1, 2, 3, 4, \ldots\} \qquad \Longrightarrow \qquad \boxed{A \triangle B = \mathbb{Z}^+}$$

(c) $\overline{C}$**:** All integers ≥ 10: $\boxed{\overline{C} = \{10, 11, 12, \ldots\}}$

(d) $A \setminus C$**:** Positive evens that are ≥ 10: $\boxed{A \setminus C = \{10, 12, 14, \ldots\} = \{2k \mid k \geq 5\}}$

(e) $C \setminus (A \triangle B)$**:** Remove all positive integers ($= A \triangle B$) from C: $\boxed{C \setminus (A \triangle B) = \{\ldots, -2, -1, 0\} = \{x}$

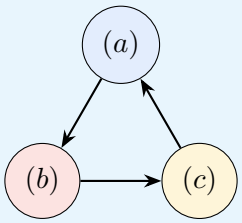
5 Problem 20 — Equivalent Characterizations of $\subseteq$

Theory Recap: Cyclic-Implication Proof Strategy

To show three conditions (a), (b), (c) are equivalent, it suffices to prove a **cycle of implications**:

$$(a) \implies (b) \implies (c) \implies (a).$$

This is more efficient than proving all six pairwise implications directly.

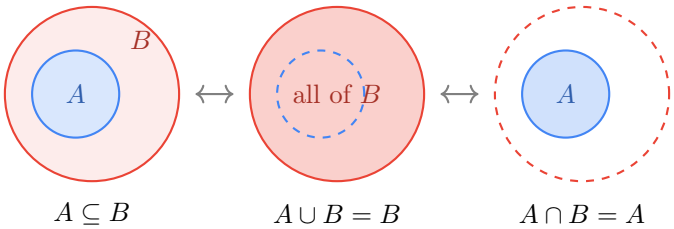


Problem Statement

Prove that for any subsets A, B of universal set U, the conditions in each group are equivalent:

1. $A \subseteq B \iff A \cup B = B \iff A \cap B = A$
2. $A \cap B = \emptyset \iff A \subseteq \overline{B} \iff B \subseteq \overline{A}$
3. $A \cup B = U \iff \overline{A} \subseteq B \iff \overline{B} \subseteq A$

Part 1: $A \subseteq B \iff A \cup B = B \iff A \cap B = A$



$(a) \Rightarrow (b)$**:** $A \subseteq B \implies A \cup B = B$.

$(\supseteq)$ $B \subseteq A \cup B$ always.

$(\subseteq)$ Let $x \in A \cup B$. Then $x \in A$ or $x \in B$. If $x \in A$, then $A \subseteq B$ gives $x \in B$. Either way, $x \in B$. $\square$

$(b) \Rightarrow (c)$**:** $A \cup B = B \implies A \cap B = A$.

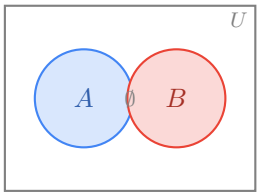
$(\subseteq)$ $A \cap B \subseteq A$ always.

$(\supseteq)$ Let $x \in A$. Then $x \in A \cup B = B$, so $x \in B$. Hence $x \in A \cap B$. $\square$

$(c) \Rightarrow (a)$**:** $A \cap B = A \implies A \subseteq B$.

Let $x \in A$. Then $x \in A = A \cap B$, so $x \in B$. $\square$

Part 2: $A \cap B = \emptyset \iff A \subseteq \overline{B} \iff B \subseteq \overline{A}$



Disjoint: $A \cap B = \emptyset$

$(a) \Rightarrow (b)$**:** $A \cap B = \emptyset \implies A \subseteq \overline{B}$.

Let $x \in A$. If $x \in B$, then $x \in A \cap B = \emptyset$—contradiction. So $x \notin B$, i.e. $x \in \overline{B}$. $\square$

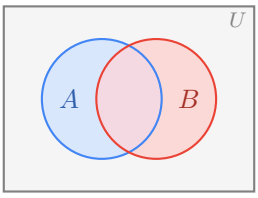
$(b) \Rightarrow (c)$**:** $A \subseteq \overline{B} \implies B \subseteq \overline{A}$.

Let $x \in B$. If $x \in A$, then $x \in \overline{B}$ (since $A \subseteq \overline{B}$), so $x \notin B$—contradiction. Hence $x \notin A$, i.e. $x \in \overline{A}$. $\square$

$(c) \Rightarrow (a)$**:** $B \subseteq \overline{A} \implies A \cap B = \emptyset$.

Suppose $x \in A \cap B$. Then $x \in B \subseteq \overline{A}$, so $x \notin A$—contradicting $x \in A \cap B$. $\square$

Part 3: $A \cup B = U \iff \overline{A} \subseteq B \iff \overline{B} \subseteq A$



Covering: $A \cup B = U$

$(a) \Rightarrow (b)$**:** $A \cup B = U \implies \overline{A} \subseteq B$.

Let $x \in \overline{A}$, so $x \notin A$. Since $x \in U = A \cup B$ and $x \notin A$, we must have $x \in B$. $\square$

$(b) \Rightarrow (c)$**:** $\overline{A} \subseteq B \implies \overline{B} \subseteq A$.

Let $x \in \overline{B}$, so $x \notin B$. If $x \notin A$, then $x \in \overline{A} \subseteq B$, giving $x \in B$—contradiction. So $x \in A$. $\square$

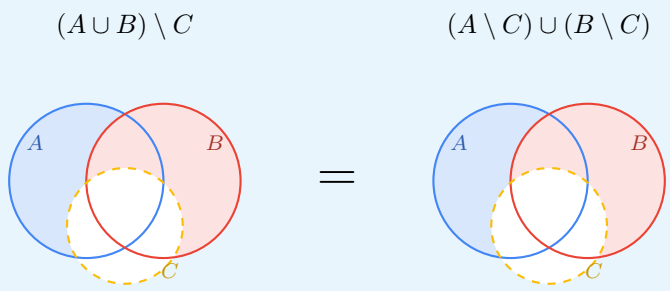
$(c) \Rightarrow (a)$**:** $\overline{B} \subseteq A \implies A \cup B = U$.

Let $x \in U$. Either $x \in B$ (then $x \in A \cup B$) or $x \notin B$ (then $x \in \overline{B} \subseteq A$, so $x \in A \cup B$). Since also $A \cup B \subseteq U$, we get $A \cup B = U$. $\square$

6 Problem 54.3 — $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$

Theory Recap: Set Difference Distributes Over Union

Mental model: "$X \setminus C$" means "start with X, **remove** everything in C."
Removing C from a union is the same as removing C from each piece separately, then taking the union of the remainders.



Both diagrams shade exactly the same region—the parts of A and B that fall outside C.

Problem Statement

Prove: $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$.

Solution

Let x be arbitrary:

$$\begin{aligned}
x \in (A \cup B) \setminus C &\iff x \in (A \cup B) \ \wedge \ x \notin C \\
&\iff (x \in A \ \vee \ x \in B) \ \wedge \ x \notin C \\
&\iff (x \in A \wedge x \notin C) \ \vee \ (x \in B \wedge x \notin C) \qquad (\wedge \text{ distributes over } \vee) \\
&\iff x \in (A \setminus C) \ \vee \ x \in (B \setminus C) \\
&\iff x \in (A \setminus C) \cup (B \setminus C).
\end{aligned}$$

Every step is $\iff$, so the sets are equal. $\square$

Logic correspondence. The key step uses the propositional law $p \wedge (q \vee r) \iff (p \wedge q) \vee (p \wedge r)$, where $p = (x \notin C)$, $q = (x \in A)$, $r = (x \in B)$.

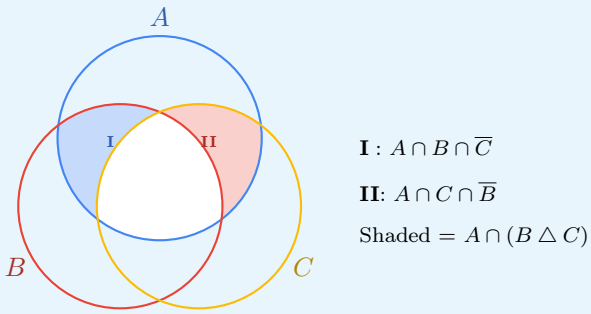
7 Problem 54.11 — $\cap$ Distributes Over $\triangle$

Theory Recap: Intersection Distributes Over Symmetric Difference

Recall: $X \triangle Y = \{x \mid x \text{ belongs to exactly one of } X, Y\}$.

The identity $A \cap (B \triangle C) = (A \cap B) \triangle (A \cap C)$ says:

> *"The part of A that sees exactly one of B, C"* equals *"exactly one of (A's overlap with B) and (A's overlap with C)."*



Algebraic analogy. Under the correspondence $\cap \leftrightarrow \cdot$ and $\triangle \leftrightarrow +$ (mod 2), the power set $(\mathcal{P}(U), \triangle, \cap)$ forms a commutative ring. This identity is the ring distributive law.

Problem Statement

Prove: $A \cap (B \triangle C) = (A \cap B) \triangle (A \cap C)$.

Solution

Let x be arbitrary. We expand both sides to the same logical form.

Left-hand side:

$$\begin{aligned} x \in A \cap (B \triangle C) &\iff x \in A \ \wedge \ x \in B \triangle C \\ &\iff x \in A \ \wedge \ \big[(x \in B \wedge x \notin C) \ \vee \ (x \notin B \wedge x \in C)\big] \\ &\iff \underbrace{(x \in A \wedge x \in B \wedge x \notin C)}_{\text{case 1}} \vee \underbrace{(x \in A \wedge x \notin B \wedge x \in C)}_{\text{case 2}} \qquad (\star) \end{aligned}$$

Right-hand side:

$$x \in (A \cap B) \triangle (A \cap C) \iff \big[x \in A \cap B \ \wedge \ x \notin A \cap C\big] \ \vee \ \big[x \notin A \cap B \ \wedge \ x \in A \cap C\big]$$

First disjunct: $x \in A \cap B$ and $x \notin A \cap C$.

- $x \in A \cap B \implies x \in A$ and $x \in B$.
- $x \notin A \cap C$ means $\neg(x \in A \wedge x \in C)$. Since $x \in A$, this forces $x \notin C$.
- Combined: $x \in A,\ x \in B,\ x \notin C$ (= case 1).

Second disjunct: $x \notin A \cap B$ and $x \in A \cap C$.

- $x \in A \cap C \implies x \in A$ and $x \in C$.

- $x \notin A \cap B$ means $\neg(x \in A \wedge x \in B)$. Since $x \in A$, this forces $x \notin B$.
- Combined: $x \in A,\ x \notin B,\ x \in C$ (= case 2).

Both sides reduce to $(\star)$, so:

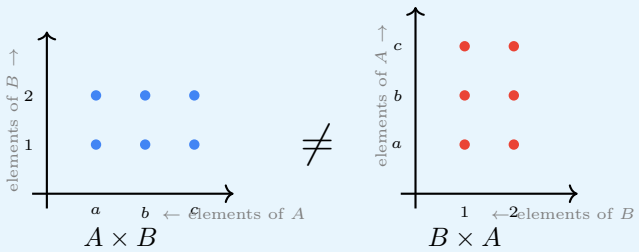
$$A \cap (B \triangle C) = (A \cap B) \triangle (A \cap C). \qquad \blacksquare$$

8 Problem 58 — $A \times B = B \times A \implies A = B$

Theory Recap: Cartesian Product

$$A \times B = \{(a, b) \mid a \in A,\ b \in B\}.$$

Order in the pair matters: $(a, b) \neq (b, a)$ when $a \neq b$.



When $A \neq B$, the grids have different shapes, so $A \times B \neq B \times A$.

Problem Statement

Prove that $A \times B = B \times A$ implies $A = B$.

Note: More precisely, $A \times B = B \times A$ implies $A = B$, or $A = \emptyset$, or $B = \emptyset$.

Solution

Assume $A \times B = B \times A$ with $A \neq \emptyset$ and $B \neq \emptyset$. We show $A \subseteq B$ and $B \subseteq A$.

$A \subseteq B$: Let $a \in A$. Since $B \neq \emptyset$, pick some $b \in B$.

$$(a, b) \in A \times B = B \times A \quad \implies \quad \text{first component } a \in B.$$

Since a was arbitrary, $A \subseteq B$.

$B \subseteq A$: Let $b \in B$. Since $A \neq \emptyset$, pick some $a \in A$.

$$(b, a) \in B \times A = A \times B \quad \implies \quad \text{first component } b \in A.$$

Since b was arbitrary, $B \subseteq A$.

By antisymmetry of $\subseteq$: $A \subseteq B \ \wedge\ B \subseteq A \implies A = B$. $\square$

Why non-emptiness matters. If $A = \emptyset$ and $B = \{1\}$, then $A \times B = \emptyset = B \times A$, but $A \neq B$. We cannot "pick $b \in B$" if $B = \emptyset$, so the argument breaks down.

9 Problem 60 — $\times$ Distributes Over Set Operations

Theory Recap: $\times$ Distributes Over $\cup$, $\cap$, $\setminus$

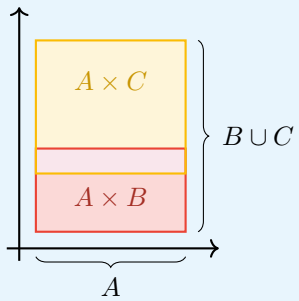
All three proofs follow the same three-step recipe:

(1) **Unpack:** $(x, y) \in A \times B \iff x \in A \wedge y \in B$.
(2) **Apply a propositional-logic law** ($\wedge/\vee$ distribution, or negation).
(3) **Repack** into set notation.

The operation in the second component "passes through" $\times$:

$$A \times (B \diamond C) = (A \times B) \diamond (A \times C) \qquad \text{for } \diamond \in \{\cup, \cap, \setminus\}.$$

Mirror identities $(B \diamond C) \times A$ work the same way on the *first* component. We prove the three non-mirror versions; the mirrors follow by swapping $x \leftrightarrow y$.



Problem Statement

Prove the following identities:

1. $A \times (B \cup C) = (A \times B) \cup (A \times C)$
3. $A \times (B \cap C) = (A \times B) \cap (A \times C)$
5. $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$

Identity 1: $A \times (B \cup C) = (A \times B) \cup (A \times C)$

Let (x, y) be an arbitrary ordered pair:

$$\begin{aligned}
(x, y) \in A \times (B \cup C) &\iff x \in A \ \wedge \ y \in B \cup C \\
&\iff x \in A \ \wedge \ (y \in B \ \vee \ y \in C) \\
&\iff (x \in A \wedge y \in B) \ \vee \ (x \in A \wedge y \in C) \qquad (\wedge \text{ dist. over } \vee) \\
&\iff (x, y) \in A \times B \ \vee \ (x, y) \in A \times C \\
&\iff (x, y) \in (A \times B) \cup (A \times C). \qquad \blacksquare
\end{aligned}$$

Identity 3: $A \times (B \cap C) = (A \times B) \cap (A \times C)$

Let (x, y) be an arbitrary ordered pair:

$$\begin{aligned}
(x,y) \in A \times (B \cap C) &\iff x \in A \ \wedge \ y \in B \cap C \\
&\iff x \in A \ \wedge \ y \in B \ \wedge \ y \in C \\
&\iff (x \in A \wedge y \in B) \ \wedge \ (x \in A \wedge y \in C) \quad \text{(idempotency of } \wedge) \\
&\iff (x,y) \in A \times B \ \wedge \ (x,y) \in A \times C \\
&\iff (x,y) \in (A \times B) \cap (A \times C). \qquad \blacksquare
\end{aligned}$$

Identity 5: $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$

Let (x, y) be an arbitrary ordered pair:

$$\begin{aligned}
(x,y) \in A \times (B \setminus C) &\iff x \in A \ \wedge \ y \in B \ \wedge \ y \notin C \\
&\iff \underbrace{(x \in A \wedge y \in B)}_{(x,y) \in A \times B} \ \wedge \ \underbrace{\neg(x \in A \wedge y \in C)}_{(x,y) \notin A \times C} \qquad (\star) \\
&\iff (x,y) \in (A \times B) \setminus (A \times C).
\end{aligned}$$

Justification of $(\star)$*:* We need $(p \wedge q \wedge \neg r) \iff (p \wedge q) \wedge \neg(p \wedge r)$, where $p = (x \in A)$, $q = (y \in B)$, $r = (y \in C)$.

$(\Rightarrow)$ $p, q, \neg r$ all hold, so $p \wedge q$ holds. Since $\neg r$, the conjunction $p \wedge r$ is false, so $\neg(p \wedge r)$ holds.

$(\Leftarrow)$ From $p \wedge q$ we have p and q. From $\neg(p \wedge r)$ and p, we get $\neg r$. $\square$

Identities 2, 4, 6 are the mirror versions $(B \diamond C) \times A = (B \times A) \diamond (C \times A)$ for $\diamond \in \{\cup, \cap, \setminus\}$. Replace "$x \in A$" with "$y \in A$" throughout—the proofs are identical.